Contact

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Top Skills

Desenvolvimento de software
Computer Vision
Time Series

Languages

Francês (Professional Working)
Inglês (Full Professional)
Português (Native or Bilingual)
Espanhol (Elementary)

Certifications

Docker: criando e gerenciando containers
Testes em .NET: criando testes de unidade com xUnit
Developing on AWS (3.4.0)
.NET: criando uma API Web com ASP.NET Core
Hugging Face: aprimorando eficiência de modelos com Optimum

Publications

MÉTODO PARA HOMOLOGAÇÃO DE SMART TV COM APRENDIZADO DE CONTEXTO PERSONALIZADO
Electrochemical Impedance Spectroscopy Applied in Theoretical Model of Exclusion Zone
Using Near Infrared Spectroscopy and Machine Learning to diagnose Systemic Sclerosis
Crônicas surreais de minha não interessante história de vida
On AutoMLs for Short-Term Solar Radiation Forecasting in Brazilian Northeast

Hugo Abreu

Senior Developer (Full Stack/Embedded) | AI Specialist | MSc Systems Engineering
Manaus, Amazonas, Brazil

Summary

I am passionate about technology and science, and I love the excitement of developing innovative all the way to the production stage. I also enjoy engaging with clients to discuss their needs, always from a technical perspective. Artificial intelligence is my true passion, and I am building my career around it. I wrote and co-wrote a some papers for journals and conferences, checkout my scholar Google profile: https://scholar.google.com/citations?user=CUtwvawAAAAJ&hl=en

Experience

Sidia Instituto de Ciência e Tecnologia

Senior Software Developer
June 2022 - Present (4 years 6 months)
Manaus, Amazonas, Brasil

- PoC development for SmartTV: Position Estimation of TVs in indoor environment to prevent piracy and Content suggestion.User behavior analysis for VoD via swarm of agencies and content retrieval. Graph network modeling for visually impaired TV navigation via remote control.

- Python/NodeJS/React: Generative AI Animations based on human movements + FastAPI & React. Pose estimation and object detection with mediapipe and yolo respectively. Video processing and similarity detection. NLP (Spacy, NLTK) for PT-BR x LIBRAS dataset generation. Using LSTM (TensorflowJS) and T5 Transformers (HuggingFace) model for Seq2Seq translation. Audio transcription with Whisper and Vosk and native libs in TizenOS. Also creating a full lib in C# from scratch for pt-BR to LIBRAS translation based on semantic and syntax structure changes from both languages. Also working to deploy the Libras translation for DTV+ (TV3.0) broadcasting.

- Developing a NodeJS + React + Electron app (front-end). Using Python for image and video description tasks with CLIP and BLIP models, and applying NLP and Transformer models for paraphrasing (HuggingFace).

- C++ (20, 17, 11), C# (.net 8): DTVPlus development for Tizen TVs. Maintaining, debugging, and adding new features to C/C++ and C# libraries and applications for Tizen, specifically for cameras and IoT devices.

- Automating firmware performance tests with a C++/Bash app for embedded Linux (TizenOS), using Python Flask for APIs and front-end rendering, Docker + PostgreSQL for the app's database.

- Applying deep learning to verify checkpoint images and automatically classify performance KPIs. Using Python Django RestAPI for data scraping, OpenCV and Scikit-learn + TensorFlow for generating classifications.

- Scientific paper and patents writing.

Outlier
Prompt Engineer
June 2023 - February 2025 (1 year 9 months)

Enhancing Large Language Models for specific training needs:

- Reviewing code generated by LLMs in multiple programming languages (Go, Python, C, C++, C#, Java, JavaScript, TypeScript) to address specific prompts.

- Specializing in data science and training large language models to solve diverse prompt-related challenges.

- Working on reinforcement learning from human feedback projects related to prompt generation, code generation, code review, and code fixing for large language models.

- Applying adversarial techniques in chatbots for malware generation.

- Detecting malware generated by LLMs through code reviews and analysis of prompt intent.

- Generating new prompts from zero to test the capabilities of the LLM and reviewing prompts generated by others to adjust in terms of usability and difficulty levels.

- Reviewing labeling generated by others labelers as a trusted prompt engineer.

FITec
Systems Engineer
February 2019 - May 2022 (3 years 4 months)
Greater Recife

- Customizing Debian based Linux distribution for OEM installation.

- Automating the massive import of network data into VitalQIP, while maintaining network structure and inputting end-client information.

- Solving multidisciplinary problems using machine learning, optimization, statistics, and other mathematical and computational tools, including digital signal and image processing, with Python libraries such as NumPy, Scikit-learn, SciPy, and Pandas.

- Developing web applications (APIs) and microservices with computational models using Spring Boot and Flask (Python). Utilizing the Selenium testing tool (Python version) for web scraping.

- Analyzing and correcting security vulnerabilities in RHEL systems, including hardening of servers.

- Developing C# .Net applications for Windows CE and Windows, used for supervising and adjusting train routes, including communication and system integration.

- Developing quality control and automation tools for PC manufacturers using BASH and Python (FastAPI).

- Deploying Nokia NSP for router management on RHEL servers.

- Implementing Nokia AAA (RADIUS protocol) on CentOS/RHEL servers.

- Providing solutions for IP networks, including logical upgrades to network infrastructure, and ensuring the consistency of IP pools and network topology.

- Deploying and upgrading Nokia VitalQIP IPAM (IP Address Management) and DNS servers on RHEL servers.

Universidade Federal de Pernambuco
CLP lectures Assistant
August 2018 - December 2018 (5 months)
Greater Recife

- Assisting with PLC lectures for Electrical Engineering and Control & Automation courses at UFPE.

- Preparing lectures and solving exercises involving the use and programming of SIEMENS S7-1200 PLCs.

- Industrial instrumentation using stepper motors, AC & DC motors, optical encoders, and variable-frequency drives.

FACEPE - Fundação de Amparo à Ciência e Tecnologia do Estado de Pernambuco
Research Internship
August 2017 - July 2018 (1 year)
Greater Recife

- Developed near-infrared imaging equipment for diagnosing skin lesions.

- Involved in mechanical prototyping using SolidWorks, designing photoelectronic circuits, and conducting image and signal processing with Python and MATLAB.

- Performed digital processing of skin infrared light absorbance spectra, utilizing PCA and SVM for diagnosing rare autoimmune diseases.

- Developed an imaging device using infrared LEDs and photosensors. Conducted digital image processing of bovine teeth to study radiation and infrared scattering, and to assess image quality.

- Developed a LabView-based supervisory system for characterizing aqueous samples, communicating via OPC with MATLAB and metrology hardware from National Instruments, MyDAQ.

WT Engenharia
Internship
October 2017 - May 2018 (8 months)
Recife, Pernambuco, Brazil

- Maintenance and assembly of embedded electronic RF communication module for dew sensing, as a secure protocol, to working with energized high voltage transmission lines.

- IoT project applied to actuation of disconnect switches using microcontrollers (Arduino) and WIFI network (ESP8266), TCP / IP master-slave protocol.

Education

Universidade de Pernambuco
MSc. eng, Systems Engineering · (March 2020 - January 2022)

Universidade de Pernambuco
Specialist, Inteligência Artificial · (2019 - 2019)

Universidade Federal de Pernambuco
Bachelor of Engineering - BE, Mechatronics, Robotics, and Automation Engineering · (2013 - 2018)

ENSEEIHT - Ecole Nationale Supérieure d'Electrotechnique, d'Electronique, d'Informatique, d'Hydraulique et des Télécommunications
Échange, Engenharia Eletrônica · (2015 - 2016)